```java
import java.util.Scanner;
public class UserInfoLab {
    public static void main(String[] args) {


        // Part 1
        String firstName;
        String lastName;
        String passWord;
        String creditCard;


        // Create a Scanner for keyboard input
        Scanner scan = new Scanner(System.in);


        // Ask the user to enter their first and last name and pass these
        System.out.println("Enter your first name: ");
        firstName = scan.nextLine();


        System.out.println("Enter your last name: ");
        lastName = scan.nextLine();


        // values to the generateUsername method and save the returned result.
        System.out.println(generateUsername(firstName, lastName));




        // Part 2
        // Ask the user to enter a password and pass this value to the validatePassword method.
        System.out.println("Enter a password: ");
        passWord = scan.nextLine();

```

```
37
38          // The validatePassword method will check if
    the password meets the criteria:
39          System.out.println(validatePassword(passWord
    ));
40
41
42          // Part 3
43          // If the user entered a valid password in
    step 2, then ask the user to enter their
44          // credit card number and pass this value to
    the maskCreditCard method.
45          System.out.println("Enter your credit card
    number: ");
46          creditCard = scan.nextLine();
47
48
49          System.out.println(maskCreditCard(creditCard
    ));
50
51
52          // Part 4
53          // If the user entered a valid password AND
    valid credit card number, display the output
54          // as shown in the demo video
55          // https://drive.google.com/file/d/
    1sMOw5wkOgSfuUcvQhFyZ5flnv_d9qQd3/view?usp=sharing
56
57
58      }
59
60
61      public static String generateUsername(String
    firstName, String lastName) {
62          // Fill in this method and return an
    appropriate username
63          String result = "";
64          if(firstName.length() <= 2){
65              result += firstName;
66          }
67          if(lastName.length() <= 2){
```

```
68              result += lastName;
69          }
70          else{
71              result = firstName.substring(0,3) +
    lastName.substring(0,3);
72          }
73          return result;
74      }
75      public static boolean validatePassword(String
    password) {
76          String allUpperCase = "
    ABCDEFGHIJKLMNOPQRSTUVWXYZ";
77          boolean hasUpperCase = false;
78          if(password.length() < 8){
79              System.out.println("Password is less
    than 8 characters.");
80          }
81          for(int i = 0; i < password.length(); i++){
82              for(int j = 0; j < allUpperCase.length
    (); j++){
83                  if(password.substring(i, i +1).
    equals(allUpperCase.substring(j,j+1))){
84                      hasUpperCase = true;
85                  }
86              }
87          }
88
89
90          if(hasUpperCase){
91              System.out.println("Password contains an
     upper case letter");
92          }
93          else{
94              System.out.println("Password doesn't
    contain an upper case letter. ");
95          }
96          if(containsDigit((password))){
97              System.out.println("Password contains a
    digit");
98          }
99          else{
```

```
100             System.out.println("Password doesn't
    contain a digit");
101         }
102 
103 
104         if(password.length() < 8 || !hasUpperCase
     || !containsDigit((password))){
105             return false;
106         }
107 
108 
109         return true;
110     }
111     public static String maskCreditCard(String
    creditCardNumber) {
112         // Fill in this method and if the credit
    card is valid, return a masked CC
113         String result = "";
114         if(creditCardNumber.length() == 16){
115             if(allDigits(creditCardNumber)){
116                 for(int i = 0; i < 3; i++){
117                     result += "****";
118                 }
119                 result += creditCardNumber.substring
    (12,16);
120             }
121             else{
122                 System.out.println("N/A");
123             }
124         }
125         else{
126             System.out.println("N/A");
127         }
128         return result;
129     }
130 
131 
132     /**
133      This method verifies that the string contains
    at least one numeric digit
134      @param str The string to check
```

```
135      @return true or false if a digit is present
136      */
137     public static boolean containsDigit(String str
    ) {
138         char[] chars = str.toCharArray();
139         for (char c: chars) {
140             if (Character.isDigit(c))
141                 return true;
142         }
143         return false;
144     }
145 
146 
147     /**
148      * Checks if the entire String is all numerical
149      * @param str The string to check
150      * @return true or false if the string is ALL
    digits
151      */
152     public static boolean allDigits(String str) {
153         char[] chars = str.toCharArray();
154         for (char c: chars) {
155             if (!Character.isDigit(c))
156                 return false;
157         }
158         return true;
159     }
160 
161 
162 }
163 
164 
165 
```